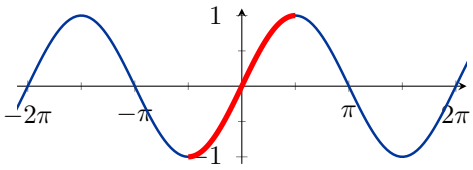
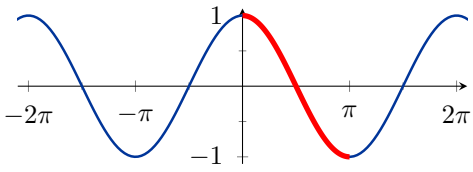
6.3: Inverse Trigonometric Functions

Note that $\sin(x)$ and $\cos(x)$,

- take in the measure of an angle, and
- return the ratio of the sides.

Recall that trigonometric functions are periodic:





Consider the following restricted domains and associated ranges:

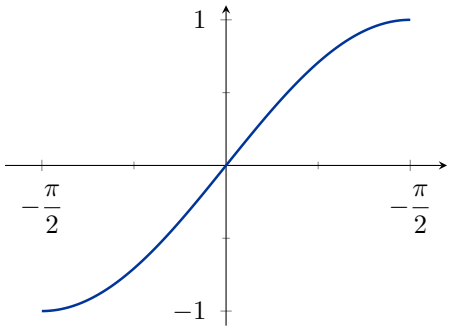
	Domain	Range
$y = \sin(x)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$[-1, 1]$
$y = \cos(x)$	$[0, \pi]$	$[-1, 1]$
$y = \tan(x)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$(-\infty, \infty)$

When we use the restricted domains, then these functions are one-to-one. The resulting inverse functions are:

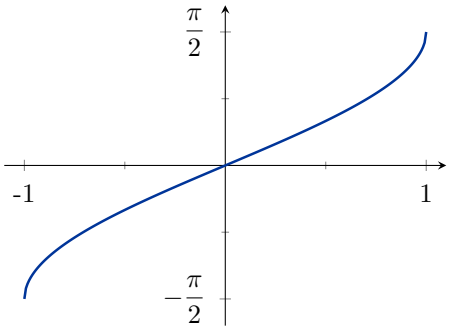
$$\arcsin(y) = \sin^{-1}(y)$$
$$\arccos(y) = \cos^{-1}(y)$$
$$\arctan(y) = \tan^{-1}(y)$$

Example. Since $\sin\left(\frac{5\pi}{12}\right) \approx 0.96593$, what is $\sin^{-1}(0.96593)$?

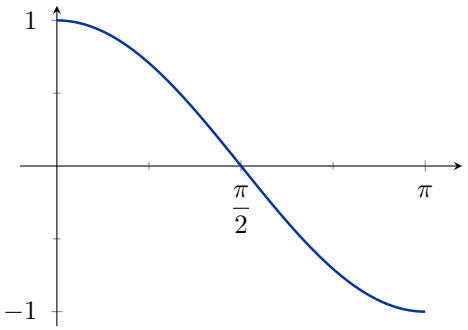
$y = \sin(x)$



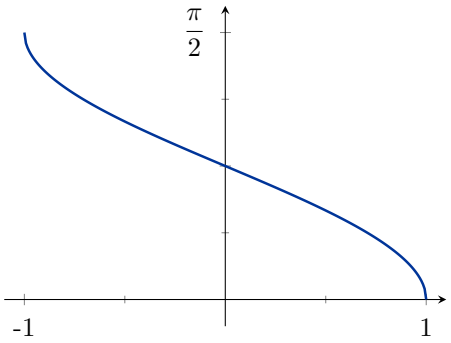
$y = \sin^{-1}(x)$



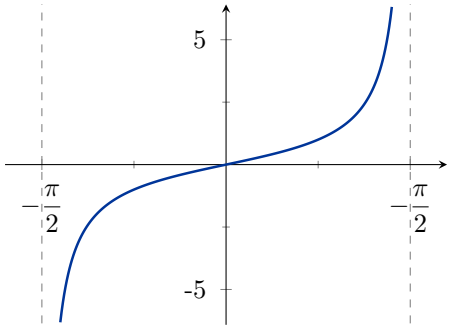
$y = \cos(x)$



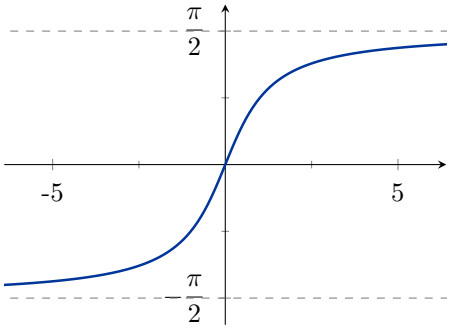
$y = \cos^{-1}(x)$



$y = \tan(x)$



$y = \tan^{-1}(x)$



Example. Evaluate the following:

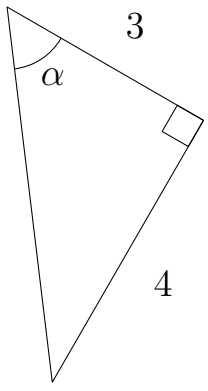
$\sin^{-1}\left(\dfrac{\sqrt{2}}{2}\right)$

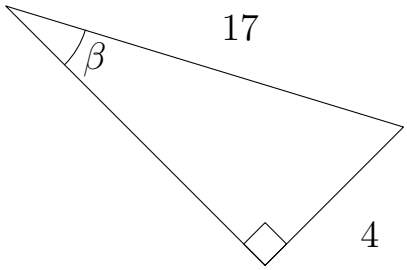
$\cos^{-1}\left(-\frac{1}{2}\right)$

$\tan^{-1}(-1)$

$\tan^{-1}\left(\dfrac{\sqrt{3}}{3}\right)$

Example. Use inverse trigonometric functions to solve for the following angles:





Example. Evaluate the following:

$\sin^{-1}\left(\sin\left(37^\circ\right)\right)$

$\cos^{-1}\left(\cos\left(\frac{9\pi}{4}\right)\right)$

$\sin\left(\cos^{-1}\left(\frac{40}{41}\right)\right)$

$\tan\left(\sin^{-1}\left(\frac{3}{5}\right)\right)$

Example. Simplify the following:

$\cos\left(\sin^{-1}\left(\frac{x}{3}\right)\right)$, where $-3 \leq x \leq 3$

$\sin\left(\tan^{-1}\left(4x\right)\right)$